

Emmanuel Owusu Asante

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SUMMARY

Machine-learning practitioner with an MPS in Analytics (Machine Intelligence) and a mathematics background. Hands-on work across computer vision, clinical predictive modeling, and reproducible data analysis in Python.

EDUCATION

Northeastern University

2025

Master of Professional Studies, Analytics — Concentration: Machine Intelligence

University of Mines and Technology (UMaT), Ghana

BSc, Mathematics

TECHNICAL SKILLS

Languages Python, MATLAB, SQL

ML & CV PyTorch, scikit-learn, CNNs, semantic segmentation, transformers (SegFormer / ViT), OpenCV / scikit-image

Data pandas, NumPy, matplotlib, seaborn, Jupyter, cross-validation, EDA, feature engineering

Tools Git / GitHub, ImageJ / Fiji, PyImageJ, Linux

SELECTED PROJECTS

LightCSPNet — Cloud Detection & Segmentation, Landsat 8

2026

Independent research · Python, PyTorch

- Built a compact CNN encoder–decoder for pixel-level cloud segmentation, progressing from USGS EarthExplorer scenes and Otsu pseudo-masks to the annotated 38-Cloud benchmark.
- Reported IoU 0.73 / Dice 0.85 — ahead of a CLIP patch classifier and zero-shot SegFormer / ViT baselines.

Spinal Abscess Prediction from ICU Records

2024

Northeastern University, EAI 6010 · team of 2 · Python, scikit-learn

- Constructed a case/control cohort from the MIMIC-III critical-care database, joining admissions, diagnoses, procedures, and patient tables.
- Trained and evaluated a logistic-regression classifier with stratified 5-fold cross-validation: AUC 0.776 (95% CI 0.71–0.87), sensitivity 0.69, specificity 0.65.

Particle-Size Analysis with ImageJ & PyImageJ

2024

Research collaboration · Python, PyImageJ / Fiji

- Scripted headless Fiji from Python to batch-segment metal-powder micrographs (Ti-6Al-4V, ball-milled powders) and derive size distributions, with downstream regression comparing measurement methods.

NearPeer Enrollment Analysis

2024

Northeastern University, ALY 6080 · team of 4 · Python, pandas

- Merged platform-engagement data with university records; cleaned, joined, and engineered time-based features for exploratory analysis of engagement vs. enrollment.

MCQ Marking Software

MATLAB · Image Processing Toolbox

- Desktop GUI that auto-grades scanned multiple-choice answer sheets against a marking scheme using thresholding and connected-component analysis.

EXPERIENCE

Curriculum & Autograding — iZen AI Academy

- Author Python, NumPy, pandas, and introductory machine-learning course materials and GitHub Classroom autograders used across multiple cohorts.